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Individual Activity 4: Design Your Own System Using UML

Part 1: System Idea & Recall

1. UML Diagrams (Descriptions)

1. **Use Case Diagram**
   Shows the interaction between users (actors) and the system, focusing on what the system does from the user's perspective.
2. **Activity Diagram**
   Represents the step-by-step workflow of a process, showing decisions, actions, and flow of control.
3. **Class Diagram**
   Displays the structure of the system by showing classes, their attributes, methods, and relationships.
4. **Sequence Diagram**
   Illustrates how objects interact in a sequence over time, focusing on message flow between components.

2. System Idea

System Title:

Online Food Ordering System

Short Description:

The Online Food Ordering System allows customers to browse menus, select food items, and place orders through a digital platform. Users can create accounts, log in, and track their orders in real-time. The system also enables restaurants to manage their menu and receive customer orders efficiently. It improves convenience for customers and streamlines food service operations.

Main Features:

- User registration and login
- Browse food menu
- Add items to cart
- Place and track orders
- Online payment option

Part 2: UML Diagram Creation

1. Use Case Diagram (Text Representation)

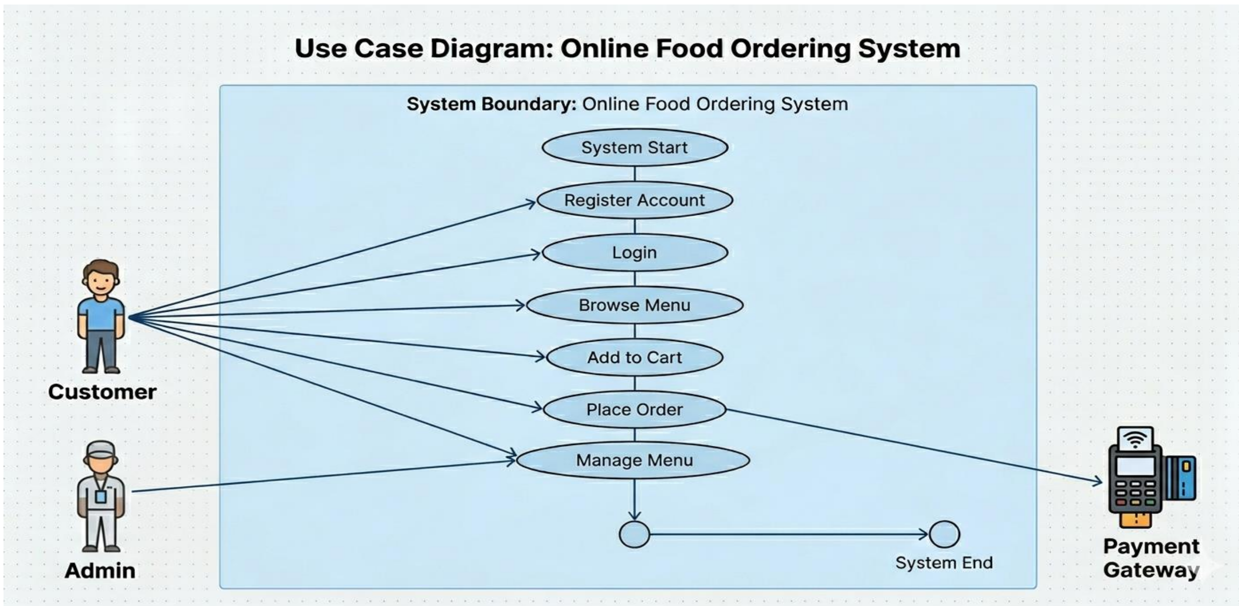
Actors:

- Customer
- Admin

Use Cases:

- Register Account
- Login
- Browse Menu
- Add to Cart
- Place Order
- Manage Menu (Admin)

USE CASE DIAGRAM: ONLINE FOOD ORDERING SYSTEM



2. Activity Diagram (Step-by-Step Process)

Part 3: SDLC Application

Planning

Identify the purpose of the system, target users, and required features such as ordering, payment, and tracking.

Design

Create UML diagrams, database structure, and user interface layout to visualize how the system will function.

Implementation

Develop the system using programming languages and frameworks, integrating features like login, ordering, and payment.

Testing

Check for errors, bugs, and system performance to ensure all features work correctly before deployment.

Part 4: Reflection

1. What did you learn from creating UML diagrams?

I learned how UML diagrams help visualize system processes and structure, making it easier to understand how different parts of a system interact.

2. Why is planning (SDLC) important before building a system?

Planning is the first thing you need to do, it is important because it ensures the system is well-organized, reduces errors, and helps developers clearly understand what needs to be built.